Plotline: technical brief

Live figures and deep dive: collaboration95.github.io/hr-2026/brief · Evidence and rerun commands: /brief/#reproduce · Code: github.com/Collaboration95/hr-2026

Plotline stores **the decision, not the data**. It reduces a multi-layer, multi-year forest stack to the three facts an EUDR plot screen reads, then **proves for every cell** that the reduced layer gives the same answer as the original layers. A **6,146 B** decision layer replaces **623,237 B** of source rasters (99.0% smaller), **744,732 / 744,732** cells decode identically, registered plots re-check **44×** faster (83× without NumPy, 410× from a content-bound answer table) and new boundaries **5.2×**.

4 GeoTIFFs `623,237 B` → **drop unread facts** `209,095 B` → **20 codes / cell** `JRC at Hansen centre` → **split store** `6,146 B` → **centre-rule scan** `exact GEOS at edges` → **20-bin histogram** `one reducer → answer`

1 · Problem and inputs

→ framing

- **Problem.** EUDR screening is a spatial join that keeps re-running: plot polygons × forest state on 31 Dec 2020 × loss since, repeated whenever a layer updates. The screen asks Q1 *forest on the cut-off?*, Q2 *that forest lost since?*, and Q3 (the supplier's own check) *any Hansen loss in 2021–23?*
- **Layers.** Hansen GFC 2025 v1.13 `treecover2000`, `lossyear`, `datamask` (tile 10N_090E, 0.00025° ≈ 27.5 m) and JRC Global Forest Cover 2020 v4 (≈ 9.2 m; 3 × 3 JRC cells nest exactly in each Hansen cell). No resampling.
- **Region and stack.** Surat Thani, Thailand: 1,233 × 604 = 744,732 cells, 17 supplied plots. Python 3.14 engine (numpy, rasterio, shapely 2.1.2 / GEOS); React + TypeScript + Vite review app.
- **AI.** No model in the pipeline. **Performance work ran as verification loops. An independent validator agent, tasked only with breaking results**, re-derived them with its own code and exact rational arithmetic. A learned codec selector was built and rejected: a plain optimised GeoTIFF beat it by 10,217 B.

2 · Schema: keep the decision

→ layers, schema

- **Layer analysis before any codec.** `treecover2000` is **72% of the Hansen GeoTIFF bytes** and no question reads it. Loss years 2001–2020 and JRC sub-cell detail do not change any answer either. All three are dropped.
- **Code space.** One of **20 codes** per Hansen cell: unknown, water, or land × {2020 non-forest, 2020 forest, forest state unknown} × {no post-2020 loss, 2021 … 2025}.
- **Two levels.** *Screening* (`eudr-screen.v1`, 38,513 B) answers Q1–Q3. *Decision* (`eudr-decision.v1`, 6,146 B) answers Q1–Q2 and forgets the loss year on land that was not forest in 2020, merging codes 3–7 into code 2. The lossless store (535,012 B) is the audit tier for re-derivation.

3 · Store format

→ store

- `.plds` **layout.** 20 B prefix (`PLDECID1`, version, header length, header CRC32) · 675 B canonical-JSON header (grid, level, encoding, sha256 of the four source rasters, the payload and the decoded codes) · 5,451 B payload.
- **Split encoding.** Four 1-bit planes (unknown, water 449 B, 2020 forest 4,453 B, forest-state unknown 36 B), then post-2020 loss as **LEB128 gaps between ranks among forest cells** (366 B) and one year symbol per event (145 B). Each stream keeps the smallest of 10 compressor settings. The result is **28% smaller than the best GeoTIFF** of the same codes (8,532 B) and beats whole-raster zstd, xz, LZMA2 and RLE.
- **Integrity.** CRC before parse, payload hash, bounded decoders, then the decoded-raster hash. The decoder works on packed planes and never expands the 744,732-cell raster.

4 · Query path

→ query, paths, phases

- **Centre rule, exact.** A cell counts when the polygon covers its centre (boundary-inclusive, holes excluded). A row scan settles clear centres, and **only ambiguous centres go to GEOS** `covers`. The codes become a 20-bin histogram, and one shared reducer produces the answer. Unknown is never reported as clear.
- **Registered plots.** Each plot's cell list is stored once. A re-check is then a bincount (**44×**), or **83× in a numpy-free path** that decodes only the registered cells from the packed planes. **An answer table bound to plot ID, geometry hash and store hash** serves unchanged plots in 0.2 µs (**410×**) and misses as soon as either side changes.
- **New boundaries.** Pure-Python scan for small windows and NumPy for large ones, with the switch fitted on a 224-shape sweep: **5.2×**. A batch evaluator handles 500 polygons at 146,000/s (15.8×).
- **Phases 1–4** changed no output or stored byte: registered index 24× → 44×, new boundaries 2.2× → 5.2×.

5 · Proof and benchmarks

→ benchmarks

- **Every cell, therefore every plot.** Each output depends only on the selected cells' codes, so **exhaustive cell equality covers every possible boundary in the region**, not just the plots tested. **744,732 / 744,732** cells match the original GeoTIFFs through an independent path at both levels. Also: 17 / 17 plots on every field, 9,829 / 9,829 fuzzed polygons, 0 fast-vs-reference mismatches on 4,920 validator geometries, and one-cell-flip, moved-vertex and flipped-JRC-bit controls all caught.

17 supplied plots	Hot	Load+query	Peak
Original layers (P0)	1.44 ms	11.1 ms	75 MB
Decision layer, fast select (P4)	0.27 ms · **5.2×**	1.5 ms	40 MB
Registered index (P5d)	0.032 ms · **44×**	1.3 ms	37 MB
Numpy-free re-check (P5Ld)	0.017 ms · **83×**	1.0 ms	23 MB
Answer table (P6d)	0.0035 ms · **410×**	0.24 ms	21 MB

- **Storage at tile scale.** One whole 10° tile (1.6 billion cells) shrinks from 421.9 MB to **8.2 MB**, 94.5% below a zstd-19 recompression of the originals.
- **Method.** Isolated child processes and two full runs (median spread 0.5%). Every timed answer is bound to the proof's digests. Apple M5, warm cache.

6 · Findings and rejected paths

→ decisions

- **Single-layer screens raise the wrong alarm.** Hansen alone flags 7 of 17 plots for post-2020 loss. Against JRC 2020 forest only P-0001 remains, with **2 of its 38 loss cells** (2022) on mapped 2020 forest. The supplied declarations do not reproduce from the layer they name.
- **The validator broke real things.** It found an unprotected header, an `OverflowError` path, level mix-ups, allocation amplification from crafted input and missing content binding. Each was fixed behind a test that failed on the old code.
- **Measured and dropped.** GPU (launch ≈ the whole 16 µs query; no float64 on Metal), CPython JIT (2–3% slower), Cython (≈ 6%), numba (0.2–0.7 s compile), PyPy (no wheels), and per-plot histogram caching (it only games the benchmark).

7 · Design limits

→ limits

- **All-or-nothing edge cells.** Answers are counts of ≈ 0.076 ha cells, so partial edge coverage and the 0.5 ha minimum area are not modelled.
- **JRC is sampled once per Hansen cell.** Centre, 3 × 3 majority and any-forest give the same decision for all 17 plots, but only one plot has mapped forest.
- **The layer is rule-specific.** A changed rule or a new layer (e.g. protected areas) means re-deriving from the audit tier: seconds per region, ≈ 7 CPU-minutes per tile (estimated).